

Qualification

Thapar Institute of Engineering and Technology

B.Tech in Computer Engineering — CGPA: 8.61/10

2024 – 2028 (Expected)

Patiala, India

Modern Senior Secondary School

Class XII – PCM with Computers — Percentage: 92.6%

2022 – 2024

Patiala, India

Technical Skills

Languages: C++, Python, JavaScript, HTML, CSS, MATLAB, R

Frameworks/Libraries: React.js, Node.js, FastAPI, MediaPipe, dnd-kit, TensorFlow, OpenCV, Pandas, NumPy, scikit-learn

Cloud / Databases: MongoDB, Firebase, SQL, Netlify, Render, Supabase, Vercel

Developer Tools: Git, Docker, Linux, Postman

Coursework: Data Structure and Algorithm, Operating Systems, Relational Database Concepts, Object Oriented Programming, Artificial Intelligence, Computer Networks

Experience

Auraflo (TIET Ventures Lab)

Mar 2025 – Sep 2025

AI/ML Intern

Remote

- Built a low-latency computer vision pipeline utilizing TensorFlow Lite and OpenCV to process **real-time video streams at 45 FPS**, optimizing model inference for edge devices.
- Architected an automated state-transition algorithm using confidence-based filtering, achieving a **95% execution accuracy across 500+ live testing sessions**.

Projects

AtomQuest — Compliance Portal | *Next.js 14, React, Supabase, Vanilla CSS* (GitHub | Live)

May 2026

- Optimized product architecture by implementing a serverless-safe caching layer (Next.js 14), reducing processing overhead by **60%** and improving the performance of critical data workflows.
- Engineered a workflow-driven compliance escalation engine featuring **end-to-end client-side encryption**, ensuring strict data security and safeguarding sensitive user information.

Spatial Console — Gesture-Controlled Structural Simulation | *React, Three.js, MediaPipe* (GitHub)

Feb. 2026

- Engineered a spatial-partitioning collision system **3×3 grid locality** for a **custom voxel physics engine**, reducing narrow-phase collision checks by **95–98% across benchmark scenes** (modern house, old house, multistory).
- Built a browser-native **3D voxel engine** with real-time **MediaPipe gesture tracking** and a **first-person walkthrough mode**; maintains **150+ FPS** with a **457 KB** gzip production bundle.

Code Vault — Competitive Programming Platform | *React, Node.js, MongoDB, Tailwind CSS* (GitHub | Live)

Jan. 2026

- Deployed a real-time evaluation platform handling **30+ concurrent teams**, building an automated anomaly detection system with **tab-switching trackers** to prevent malicious submissions.
- Designed a low-latency backend infrastructure (Node.js, MongoDB) with **2-second auto-refresh** and **3-second leaderboard polling**, optimizing data synchronization for live evaluation workflows.

Positions of Responsibility

LEAD Society (TIET)

Oct 2024 – Present

Technical Secretary

Patiala, India

- Built the official LEAD Society website: lead2025.netlify.app using HTML, CSS, and JavaScript; maintained the site and deployment.
- Mentored **100+** members in web development and Git workflows; spearheaded technical events for a **500+** student community.

Modern Senior Secondary School

2022 – 2023

Vice Art & Cultural Captain

Patiala, India

- Organized 4 **school-wide** events and anchored 3 ceremonial programs, reaching a cumulative audience of **2,000+** students and faculty.

Achievements

- **Merit III** Scholarship awarded for scoring **95.4** percentile in JEE Mains.
- Specialist @Code360, Solved **500+** challenges (LeetCode, GFG) emphasizing low-latency execution (Graphs, Trees, DP).
- **District Gold Medalist & School Chess Champion:** Recognized for excellence in competitive strategy and performance.